JADE RYAN BALABA

Licensed Mechanical Engineer | E-Systems & Wiring Harness Product Engineer

Cebu, Philippines (open to relocation) · +63 997 613 7069 · jaderyan.balaba@gmail.com · linkedin.com/in/jade-ryan-balaba-642645149

PROFESSIONAL SUMMARY

Licensed Mechanical Engineer with over 4 years of experience in automotive wire harness (EDS) engineering, release readiness and continuous improvement. Owns 2D/3D harness drawings, schematics, BOMs and change control for major OEM vehicle programs. Builds Python and Power Platform tools that automate engineering document validation, cutting manual review time by over 60% and catching release defects before sign-off. Completing an M.S. in Management Engineering with a thesis on AI-assisted document validation.

KEY RESULTS

- **2.59× savings vs. goal:** delivered 259% of the annual continuous-improvement savings target as CIP Leader (2025), with 142% of the quantity target and 103% of the quality target.
- **27 proposals implemented** out of 36 submitted across engineering units.
- **60%+ less manual review time** on harness drawings through custom Python extraction and verification tools.
- **Company AI team member:** endorsed by management to build AI readiness training and automation frameworks.

PROFESSIONAL EXPERIENCE

Product Engineer I — Harness Owner & CIP Leader

2023 – Present

Lear Corporation, Philippines Engineering Branch · Cebu, Philippines

- **Harness ownership & release readiness:** Full engineering ownership of major OEM wire harness programs, including fuel-system and passenger-seat harnesses. Produce and check 2D/3D drawings, schematics, electrical sketches (ESK) and vehicle routing, and manage BOMs through to release.
- **Change management:** Execute change requests (PCR/ECR), derivative updates and drawing markups in the customer's engineering release and change-management systems.
- **Quality & risk mitigation:** Facilitate DFMEA and Failure Mode Avoidance (FMA) sessions with manufacturing, quality and purchasing teams to enforce design compliance.
- **Continuous improvement leadership:** Led the engineering team's improvement program, guiding proposals from idea through clarification and verification; achieved 259% of the annual savings target and implemented 27 proposals.
- **AI & digital transformation:** Endorsed to the company AI team; facilitate Elements of AI and AI Fundamentals sessions, launched the Automation Run framework and drive Microsoft 365 Copilot adoption.

Customer Engineer

2022 – 2023

Aurotech Corporation · Philippines

- Performed preventive and corrective maintenance on high-precision semiconductor processing equipment (dicing and grinding machinery) to strict electro-mechanical tolerances.
- Diagnosed root causes of complex equipment failures, optimized spare-parts replacement and authored technical service and compliance documentation.

Data Analyst

2020 – 2022

Office Partners 360 · Philippines

- Analyzed large unstructured operational datasets to support operational decisions and project reporting.
- Built automated dashboards and reporting models in advanced Excel and PowerPoint under tight deadlines.

ENGINEERING AUTOMATION PROJECTS

- **Automated PDF Harness Extractor (Python, PyMuPDF, Tkinter):** Desktop tool that extracts insulation, circuit, clip and splice data from complex harness drawings, color-marks matches and exports structured data to Excel. Reduced manual review time by over 60%.
- **PCR/ECR Change Verifier (Python, OCR):** Cross-checks the latest harness drawing PDF against PCR/ECR Excel change files and classifies each change as Applied, Not Applied, Needs Review or Mismatch, with a PDF review view and summary report before sign-off.
- **RFQ Harness Extraction (Python, in development):** Extracts circuit, connector, notes, part-number and route data from customer harness PDFs, with marked-up PDF and Excel export, to support derivative-aware, quotation-ready RFQ preparation.
- **Workflow automation (Power Apps, Power Automate, SharePoint):** Built a PCR approval tracker and automated attendance-log generation from Teams chat.

TECHNICAL SKILLS

- **Harness engineering & design:** 2D/3D harness schematics, physical routing, electrical sketches (ESK), BOM management, harness complexity management, manufacturing design checklists, AutoCAD, SolidWorks
- **OEM release & change management:** Engineering release and change-management systems, PCR/ECR, derivative updates, OEM program timing and build-phase terminology
- **Quality & cost:** DFMEA / PFMEA, Failure Mode Avoidance, 8D problem solving, CTO / VAVE cost validation, Lean Six Sigma principles
- **Automation & software:** Python (PyMuPDF, Tkinter, Pandas), PDF data extraction, Power Apps, Power Automate, Power BI, advanced Excel, Microsoft 365 Copilot, prompt engineering, HTML/JS, Git

EDUCATION & LICENSURE

Master of Science in Management Engineering 2024 – Present

University of San José–Recoletos

Thesis: *A Hybrid AI Framework for Product Engineering Document Validation*. Coursework: Lean Six Sigma Greenbelt, Quality Management Systems, Advanced Engineering Economy, Project Development & Management, Supply Chain Management.

Bachelor of Science in Mechanical Engineering Graduated 2021

University of Bohol

Licensed Mechanical Engineer Philippines